Simpson Querrey Institute BioNanotechnology

NORTHWESTERN UNIVERSITY

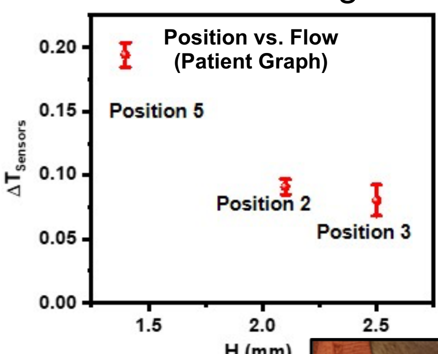
Sensitivity Analysis of Flow Rates with Varying Skin Thicknesses

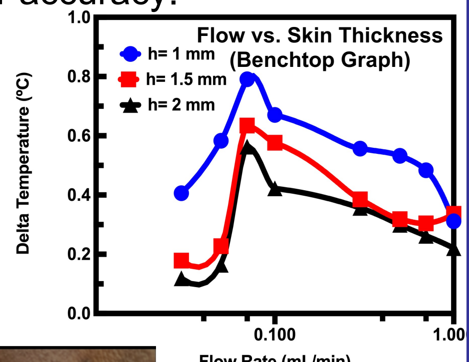
Objectives:

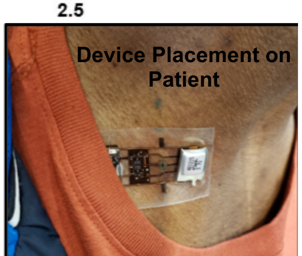
- To replicate patient data with a benchtop experiment
- To find a correlation between flow rate and skin thickness
- To identify any possible issues with the device that can be improved in the future.

Results:

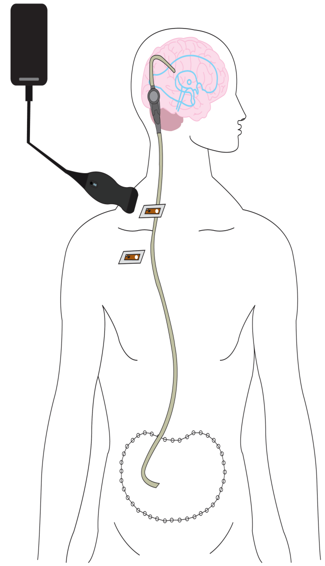
- The device is less accurate at higher flow rates.
- The thickness of the skin affects the precision of the device.
- Heat applied must be increased for more precise measurements.
- Sensor quantity must be increased or the design must be modified for greater accuracy.



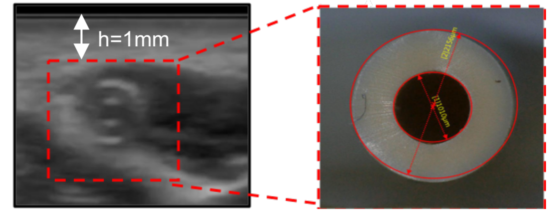




Technical Approach :



Anatomical Figure of Device Placement



Shunt Depth In Patient

R_{shunt}=2156µm

R_{cavity}=1010µm

$$k_{Sylgard^{TM}184} = 0.27\frac{W}{mK}$$

$$k_{Sylgard^{TM}170} = 0.48\frac{W}{mK}$$

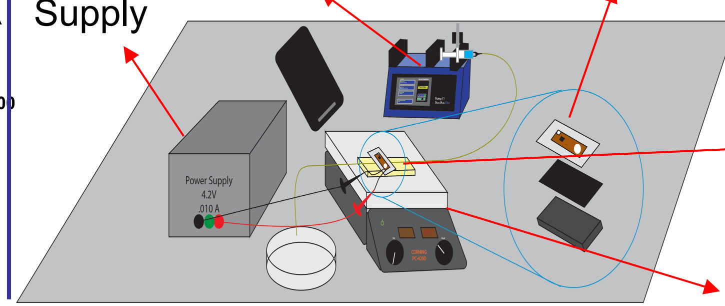
$$k_{skin} \cong 0.37\frac{W}{mK}$$

Benchtop:

Power Supply

Pump

Device 0.5mm Skin 1mm Skin



1mm Between Shunt and Skin Phantom

Heater at 45ºC

Contact: Kaan Borekci, kaan@borekci.com